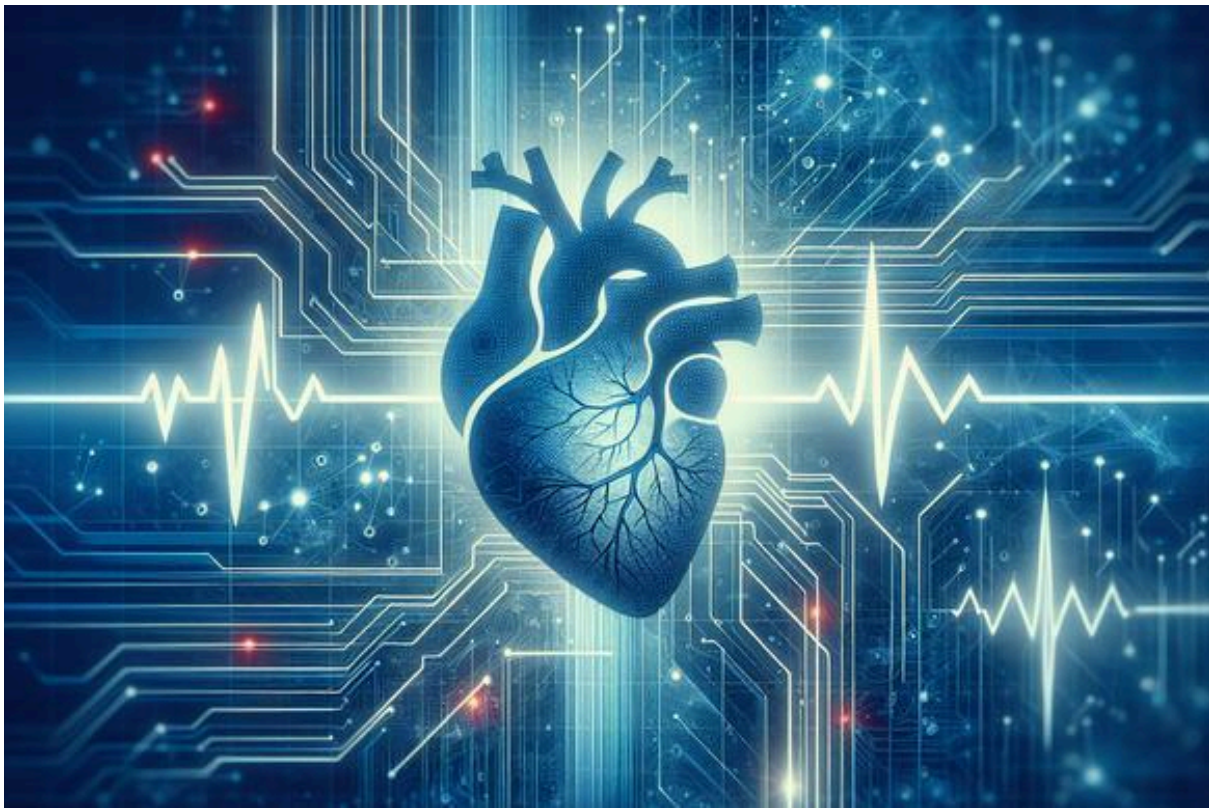
MACHINE LEARNING PROJECT REPORT

HEART DISEASE PREDICTION USING RANDOM FOREST



PROJECT BASED ON MACHINE LEARNING

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Dataset: Heart Failure Prediction Dataset (Kaggle)

ABSTRACT

Heart disease is one of the important health challenges worldwide. Machine learning techniques can help analyze health-related data and identify patterns associated with heart disease.

This project implements a Random Forest classification algorithm to predict the presence of heart disease using patient-related attributes. The Heart Failure Prediction dataset obtained from Kaggle is used for this project.

The dataset is preprocessed by handling missing values, removing duplicate records and converting categorical variables into numerical form. The dataset is divided into training and testing sets using an 80:20 ratio.

The Random Forest model is trained using 100 decision trees. Its performance is evaluated using accuracy, precision, recall and F1-score.

The model achieved an accuracy of 87.50%, precision of 88.35%, recall of 89.22% and F1-score of 88.78%.

OBJECTIVES

1. To understand and analyze a real-world healthcare dataset.
2. To perform data cleaning and preprocessing.
3. To implement the Random Forest classification algorithm.
4. To predict the presence of heart disease.
5. To evaluate model performance using classification metrics.
6. To identify important features influencing model predictions.

Keywords: Machine Learning, Random Forest, Classification, Heart Disease Prediction, Python.

1. INTRODUCTION

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data and make predictions without being explicitly programmed for every task.

Random Forest is a supervised machine learning algorithm that uses multiple decision trees to make predictions. It can be used for both classification and regression problems.

In this project, Random Forest is used to predict whether a person has heart disease based on different health-related attributes such as age, cholesterol, resting blood pressure and maximum heart rate.

1.1 Problem Statement

The main problem addressed in this project is predicting the presence of heart disease using patient health information.

The project uses a Random Forest classification model to identify patterns in the dataset and classify records into two categories:

- 0 – No heart disease
- 1 – Heart disease

1.2 Proposed Solution

The proposed solution involves collecting a public healthcare dataset, preprocessing the data, training a Random Forest classifier and evaluating its predictive performance.

1.3 Scope of the Project

This project demonstrates the application of machine learning in healthcare-related data analysis. It is developed for academic and educational purposes and is not intended to replace professional medical diagnosis.

2. DATASET DESCRIPTION

2.1 Dataset Information

Dataset Name: Heart Failure Prediction Dataset

Source: Kaggle

Dataset File: heart.csv

Format: CSV (Comma-Separated Values)

Initial Number of Records: 918

Target Variable: HeartDisease

The dataset contains patient-related health information that is used to predict the presence of heart disease.

2.2 Dataset Attributes

Attribute	Description
Age	Age of the patient
Sex	Patient's sex
ChestPainType	Type of chest pain
RestingBP	Resting blood pressure
Cholesterol	Cholesterol level
FastingBS	Fasting blood sugar
RestingECG	Resting ECG results

MaxHR	Maximum heart rate
ExerciseAngina	Exercise-induced angina
Oldpeak	ST depression value
ST_Slope	Slope of the ST segment
HeartDisease	Target variable

2.3 Target Variable

The target variable is HeartDisease.

- 0 indicates no heart disease.
- 1 indicates heart disease.

2.4 Dataset Suitability

The dataset is suitable for binary classification because it contains a target variable with two possible classes. It also contains more than 500 records, satisfying the project's dataset size requirement.

```
Dataset loaded successfully!
   Age Sex ChestPainType  RestingBP  Cholesterol  FastingBS RestingECG  MaxHR
0   40   M           ATA        140          289          0     Normal    172
1   49   F           NAP        160          180          0     Normal    156
2   37   M           ATA        130          283          0         ST     98
3   48   F           ASY        138          214          0     Normal    108
4   54   M           NAP        150          195          0     Normal    122

  ExerciseAngina  Oldpeak ST_Slope  HeartDisease
0              N      0.0       Up             0
1              N      1.0     Flat             1
2              N      0.0       Up             0
3              Y      1.5     Flat             1
4              N      0.0       Up             0
```

3. DATA PREPROCESSING.

3.1 Loading the Dataset

The dataset was loaded using the Pandas library.

```python
df = pd.read_csv('/content/dataset/heart.csv')
```

3.2 Encoding Categorical Variables

Categorical variables were converted into numerical values using one-hot encoding.

```python
df = pd.get_dummies(
    df,
    drop_first=True,
    dtype=int
)
```

3.3 Feature and Target Separation

The dataset was divided into input features (X) and target variable (y).

```python
X = df.drop('HeartDisease', axis=1)
y = df['HeartDisease']
```

3.4 Train-Test Split

The dataset was divided into 80% training data and 20% testing data.

```python
train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

Result: The dataset was prepared for training and testing the Random Forest classification model.

4. RANDOM FOREST MODEL IMPLEMENTATION

4.1 Introduction to Random Forest

Random Forest is a supervised machine learning algorithm that combines multiple decision trees to produce a final prediction.

For classification, each decision tree makes a prediction, and the forest uses majority voting to determine the final class.

4.2 Model Configuration

The Random Forest Classifier was implemented using the Scikit-learn library.

The following parameters were used:

Parameter	Value
Algorithm	Random Forest Classifier
Number of Trees	100
Random State	42
Training Data	80%
Testing Data	20%

4.3 Model Training

The model was trained using the training dataset.

```python
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)
```

4.4 Prediction

```python
y_pred = model.predict(X_test)
```

4.5 Implementation Tools

- Python Pandas Google Colab NumPy Matplotlib
- Seaborn Scikit-learn

5. RESULTS AND MODEL EVALUATION

The Random Forest model was evaluated using four classification metrics.

5.1 Evaluation Results

Metric	Score
Accuracy	87.50%
Precision	88.35%
Recall	89.22%
F1 Score	88.78%

5.2 Performance Analysis

- **Accuracy:** 87.50% of predictions were correct.
- **Precision:** 88.35% of positive predictions were correct.
- **Recall:** 89.22% of actual positive cases were identified.
- **F1 Score:** 88.78%, showing a balance between precision and recall.

5.3 Confusion Matrix

The confusion matrix displays correct and incorrect predictions, including True Positive, True Negative, False Positive and False Negative.

5.4 Feature Importance

Feature importance identifies the most influential input variables in the model.

Top 3 Features:

```
ST_Slope_Up          0.163153
MaxHR                0.119229
ST_Slope_Flat        0.113410
Cholesterol          0.107431
Oldpeak              0.099216
ExerciseAngina_Y     0.086208
Age                  0.079529
```

5.5 Result

The model achieved 87.50% accuracy, demonstrating promising performance in predicting heart disease on the test dataset.

6. CONCLUSION

The Heart Disease Prediction project was successfully implemented using the Random Forest classification algorithm.

The project involved loading a real-world dataset from Kaggle, performing data preprocessing, handling missing values, removing duplicate records and converting categorical attributes into numerical form.

The dataset was divided into training and testing sets using an 80:20 ratio. A Random Forest classifier with 100 decision trees was trained to predict the presence of heart disease.

The model achieved the following results:

- Accuracy: 87.50%
- Precision: 88.35%
- Recall: 89.22%
- F1 Score: 88.78%

The results demonstrate that Random Forest can identify useful patterns in healthcare-related data. Feature importance analysis also helped identify influential attributes.

Through this project, practical knowledge was gained in Python programming, data preprocessing, machine learning model development and performance evaluation.

Note: This project is for academic purposes and is not a clinically validated diagnostic system.

